# Vision-Language Models (VLMs)

## Large vision-language models

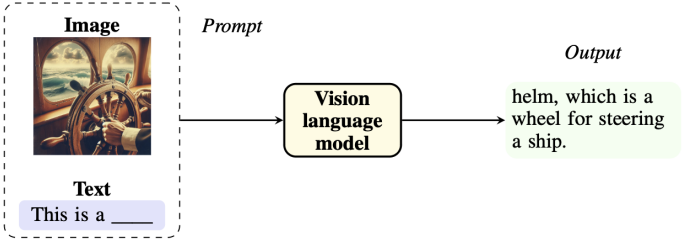
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# What Are Vision-Language Models?




- **Vision-Language Models (VLMs)** combine computer vision and natural language processing (NLP) in a single multimodal system [1, 2].
- They integrate a *vision encoder* and a *large language model (LLM)*, enabling joint reasoning over images, video, and text.
- Example tasks: image captioning, visual question answering, and image retrieval from text descriptions.

# General Overview

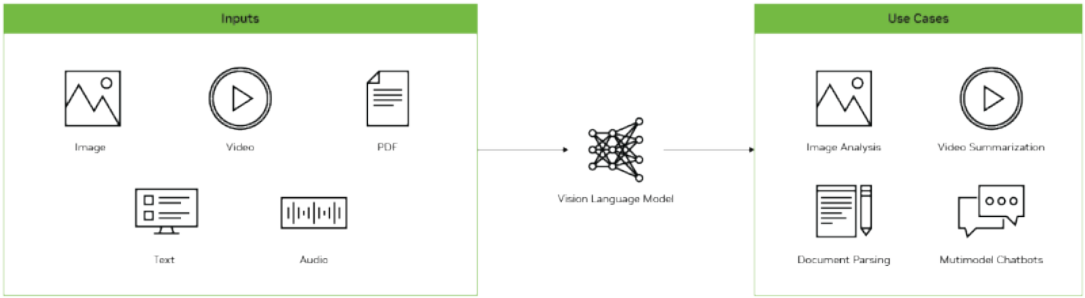



Figure: Example of a multimodal flow between vision and language [2].

# Why VLMs Matter

- They overcome the limitations of traditional vision models, since they are not restricted to a fixed set of classes or tasks.
- Thanks to training on large collections of image-text pairs, they can follow natural-language instructions and adapt to new tasks.
- They enable **multimodal generative AI** applications that integrate text, images, audio, and video.

# VLM Architecture

- Three main building blocks:
  1. **Vision encoder:** transforms images or video into embeddings.
  2. **Fusion module or projector:** aligns the vision and language spaces into a shared semantic domain.
  3. **Large language model (LLM):** interprets or generates text from the visual information.
- They are generally built on *Transformer* architectures [3], such as **CLIP** [4] or **ViT** [5].
- Integrating these modules enables multimodal reasoning and context-aware responses.

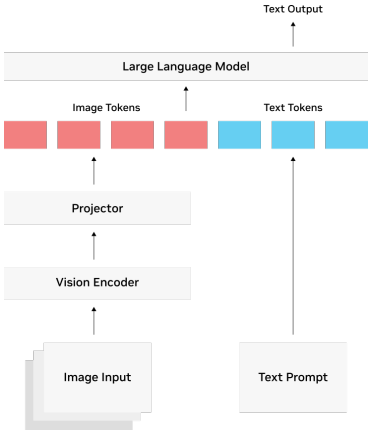
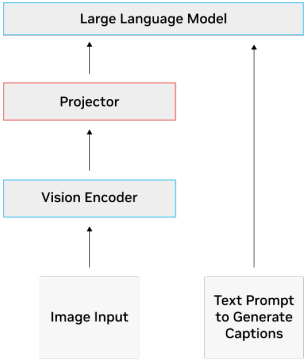



Figure: Typical architecture of a vision-language model.

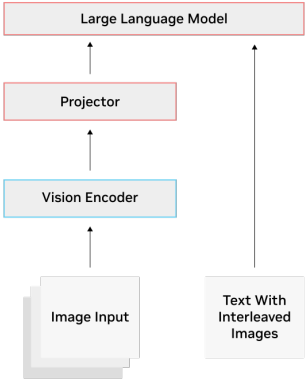
# Training VLMs

- VLMs are trained using several strategies:
  1. **Contrastive learning:** aligns image and text embeddings (e.g., CLIP [4]).
  2. **Masking:** predicts hidden parts of the text or image.
  3. **Generative training:** produces text from images or vice versa.
- They typically combine pretrained vision and language models before *fine-tuning* on task-specific multimodal data.

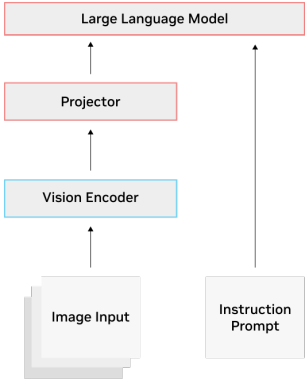
# Training Process Diagram




**Stage 1: Projector Initialization**
Trained on image caption pairs




**Stage 2: Visual-Language Pre-Training**
Trained on interleaved image-text and image-text pairs




**Stage 3: Visual Instruction Tuning**
Trained on visual language instructions

The layers of the VLM that are updated during the training stage

The layers of the VLM that are not updated during the training stage

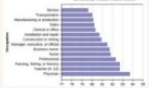
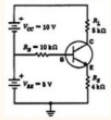
# Representative Models

- **CLIP (OpenAI)** [4]: pioneer of contrastive image-text learning.
- **GPT-4o (OpenAI)** [6]: multimodal model that processes text, images, and audio.
- **Gemini 2.0 Flash (Google)** [7, 8]: full integration of text, images, audio, and video.
- **Llama 3.2 (Meta)** [9, 10]: open model with vision and text input.
- **NVLM (NVIDIA)** [11] and **Qwen2.5-VL (Alibaba)** [12]: specialized in high-resolution images and long videos.

# Evaluating VLMs

- VLMs are evaluated with multimodal benchmarks that measure reasoning, visual understanding, and textual accuracy:
  1. **MMBench** [13], **MMMU** [14], **MM-Vet** [15]: general reasoning.
  2. **OCRBench** [16], **MathVista** [17]: understanding text and symbols in images.
  3. **VQA** [18]: answering visual questions.
- Evaluation aims to measure both accuracy and the capacity for generalization and multimodal reasoning.

# Evaluation Example

| Art & Design | Business | Science |
|---|---|---|
| **Question:** Among the following harmonic intervals, which one is constructed incorrectly?<br>**Options:**<br>(A) Major third <*image 1*><br>(B) Diminished fifth <*image 2*><br>**(C) Minor seventh <*image 3*>**<br>(D) Diminished sixth <*image 4*><br>  | **Question:** ...The graph shown is compiled from data collected by Gallup <*image 1*>. Find the probability that the selected Emotional Health Index Score is between 80.5 and 82?<br>**Options:**<br>(A) 0 (B) 0.2142<br>**(C) 0.3571** (D) 0.5<br>  | **Question:** <*image 1*> The region bounded by the graph as shown above. Choose an integral expression that can be used to find the area of R.<br>**Options:**<br>**(A)** $\int_0^{1.5}[f(x)-g(x)]dx$<br>(B) $\int_0^{1.5}[g(x)-f(x)]dx$<br>(C) $\int_0^{2}[f(x)-g(x)]dx$<br>(D) $\int_0^{2}[g(x)-x(x)]dx$<br>  |
| **Subject:** Music; **Subfield:** Music; **Image Type:** Sheet Music; **Difficulty:** Medium | **Subject:** Marketing; **Subfield:** Market Research; **Image Type:** Plots and Charts; **Difficulty:** Medium | **Subject:** Math; **Subfield:** Calculus; **Image Type:** Mathematical Notations; **Difficulty:** Easy |
| **Health & Medicine** | **Humanities & Social Science** | **Tech & Engineering** |
| **Question:** You are shown subtraction <*image 1*>, T2 weighted <*image 2*> and T1 weighted axial <*image 3*> from a screening breast MRI. What is the etiology of the finding in the left breast?<br>**Options:**<br>(A) Susceptibility artifact<br>(B) Hematoma<br>**(C) Fat necrosis** (D) Silicone granuloma<br>  | **Question:** In the political cartoon, the United States is seen as fulfilling which of the following roles? <*image 1*><br>**Option**:<br>(A) Oppressor<br>(B) Imperialist<br>**(C) Savior** (D) Isolationist<br>  | **Question:** Find the VCE for the circuit shown in <*image 1*>. Neglect VBE<br>**Answer: 3.75**<br>**Explanation**: ...IE = [(VEE) / (RE)] = [(5 V) / (4 k-ohm)] = 1.25 mA; VCE = VCC - IERL = 10 V - (1.25 mA) 5 k-ohm; VCE = 10 V - 6.25 V = 3.75 V<br>  |
| **Subject:** Clinical Medicine; **Subfield:** Clinical Radiology; **Image Type:** Body Scans: MRI, CT.; **Difficulty:** Hard | **Subject:** History; **Subfield:** Modern History; **Image Type:** Comics and Cartoons; **Difficulty:** Easy | **Subject:** Electronics; **Subfield:** Analog electronics; **Image Type:** Diagrams; **Difficulty:** Hard |

# Applications of VLMs

- **Caption generation** for images or videos.
- **Semantic visual search**: finding images from text.
- **Object segmentation and detection** with textual explanations.
- **Visual question answering (VQA)**.
- **Assisted design and image generation** (text-to-image).
- Uses in medicine, robotics, education, art, and environmental monitoring systems.

# Real-World Applications



Images



Videos



Cameras



Metadata



**Visual AI Agent**
Agent is given tasks through prompts

**Identify important activities**



Unauthorized personnel detected at 3:30pm

**Find complex activities in city**



Wrong-way Driving: Black car on wrong lane at 2pm

**Detect abnormal activities in facility**



Unauthorized Access: Potential tailgating observed at the side door

**Identify all three-pointer throws in a basketball game**



Showing clips of 4 three-pointer throws

**Detect Dangerous Situations**



Fire detected. Firefighters and two firetrucks on site.

**Create a recipe to make a pizza from the video**



3½ cups of bread flour, ¾ teaspoon active dry yeast, 1½ cups cold water, ...

# Challenges and Risks

- **Data bias:** models may reflect prejudices present in the training sets.
- **High computational costs** for training and deployment.
- **Limited generalization** to unseen or out-of-domain scenarios.
- **Hallucinations**: generating false or incorrect descriptions.
- **Governance and ethics**: the need for regulation, validation, and human oversight.

# Conclusions

- VLMs integrate visual and linguistic understanding in a single system, enabling a new generation of multimodal applications.
- They represent a step toward **generalist AI**, capable of reasoning across multiple modalities.
- Their future depends on advances in alignment, ethics, efficiency, and robustness.

# References I


[1] IBM Think (2025). *What are Vision-Language Models?* Available at: https://www.ibm.com/think/topics/vision-language-models

[2] NVIDIA Glossary (2024). *What are Vision-Language Models?* Available at: https://www.nvidia.com/en-us/glossary/vision-language-models/

[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin (2017). Attention Is All You Need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 30.

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever (2021). Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, PMLR 139, pp. 8748–8763.

[5] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*.

[6] OpenAI (2024). GPT-4o System Card. *arXiv preprint arXiv:2410.21276*.

[7] Gemini Team, Google (2023). Gemini: A Family of Highly Capable Multimodal Models. *arXiv preprint arXiv:2312.11805*.

# References II


[8] S. Pichai, D. Hassabis, and K. Kavukcuoglu (2024). Introducing Gemini 2.0: our new AI model for the agentic era. *Google Blog*. Available at: https://blog.google/technology/google-deepmind/google-gemini-ai-update-december-2024/

[9] Meta AI (2024). Llama 3.2: Revolutionizing edge AI and vision with open, customizable models. Available at: https://ai.meta.com/blog/llama-3-2-connect-2024-vision-edge-mobile-devices/

[10] A. Grattafiori, A. Dubey, A. Jauhri, et al. (2024). The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*.

[11] W. Dai, N. Lee, B. Wang, Z. Yang, Z. Liu, J. Barker, T. Rintamaki, M. Shoeybi, B. Catanzaro, and W. Ping (2024). NVLM: Open Frontier-Class Multimodal LLMs. *arXiv preprint arXiv:2409.11402*.

[12] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang, et al. (2025). Qwen2.5-VL Technical Report. *arXiv preprint arXiv:2502.13923*.

[13] Y. Liu, H. Duan, Y. Zhang, B. Li, S. Zhang, W. Zhao, Y. Yuan, J. Wang, C. He, Z. Liu, K. Chen, and D. Lin (2024). MMBench: Is Your Multi-modal Model an All-around Player? In *European Conference on Computer Vision (ECCV)*, pp. 216–233.

[14] X. Yue, Y. Ni, K. Zhang, T. Zheng, R. Liu, G. Zhang, S. Stevens, D. Jiang, W. Ren, Y. Sun, et al. (2024). MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9556–9567.

# References III


[15] W. Yu, Z. Yang, L. Li, J. Wang, K. Lin, Z. Liu, X. Wang, and L. Wang (2024). MM-Vet: Evaluating Large Multimodal Models for Integrated Capabilities. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR 235.

[16] Y. Liu, Z. Li, M. Huang, B. Yang, W. Yu, C. Li, X.-C. Yin, C.-L. Liu, L. Jin, and X. Bai (2024). OCRBench: On the Hidden Mystery of OCR in Large Multimodal Models. *Science China Information Sciences*, 67(12), 220102.

[17] P. Lu, H. Bansal, T. Xia, J. Liu, C. Li, H. Hajishirzi, H. Cheng, K.-W. Chang, M. Galley, and J. Gao (2024). MathVista: Evaluating Mathematical Reasoning of Foundation Models in Visual Contexts. In *International Conference on Learning Representations (ICLR)*.

[18] S. Antol, A. Agrawal, J. Lu, M. Mitchell, D. Batra, C. L. Zitnick, and D. Parikh (2015). VQA: Visual Question Answering. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 2425–2433.